Clinical Study Summary Report (CSSR) Document Status: Final | **Version:** 1.0 | **Date:** 04-Apr-2026

---

1. Administrative & Study Identification

Full Study Title: A Study to Assess the Efficacy and Safety of Satralizumab in Duchenne Muscular Dystrophy (DMD) **Short Title/ Acronym:** SHIELD-DMD **Protocol Number:** BN45398 / 2024-512383-65-00 **NCT Number:** NCT06450639 **Posting ID:** 39366947587872783 **Sponsor/ Company Name:** Hoffmann-La Roche **Investigational Product:** Satralizumab (Enspryng) **Phase of Development:** Phase II **Trial Status:** Recruiting **Medical Monitor:** Hoffmann-La Roche Clinical Operations / Dr. Levi Garraway (CMO)

---

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To assess the potential of satralizumab to improve bone fragility and increase bone mineral density (BMD) in boys with Duchenne Muscular Dystrophy (DMD). **Secondary Objectives:** To evaluate the effect of satralizumab on muscle function, fracture incidence, safety and tolerability, and pharmacokinetics.

2.2 Background & Rationale To address the critical unmet need for managing skeletal complications, such as bone loss and fractures, associated with DMD and long-term corticosteroid use. In DMD, the IL-6 pathway promotes inflammation in muscles and contributes to bone erosion. Blocking the IL-6 receptor with satralizumab aims to slow bone loss and muscle atrophy to help prevent fractures and improve overall muscle and bone strength.

---

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Single-arm **Blinding:** Open-label **Randomization:** N/A

3.2 Planned Enrollment & Duration Target Enrollment (N): 50 boys (Note: Study development was halted in February 2026 with <30 participants enrolled). **Number of Sites:** ~16 planned sites across 6 countries (United States, Denmark, Italy, Poland, Spain, and Ukraine). **Estimated Study Start:** Late Summer 2024. **Estimated Primary Completion:** Initially planned for 104 weeks; currently truncated to a 6-month bone mineral density collection follow-up for actively enrolled participants due to early termination.

---

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Boys aged 8 to 17 years old (with specific cohorts starting at ≥ 8 to < 12 years) at the time of screening. 2. Confirmed diagnosis of Duchenne Muscular Dystrophy (DMD). 3. Currently taking daily corticosteroids. 4. Either fracture-naïve (no prior low-trauma fractures or radiological findings of vertebral fractures) OR have a prior history of low-trauma fractures (e.g., vertebral compression or long-bone fracture). 5. Ambulatory (able to walk independently without assistive devices) or non-ambulatory.

4.2 Exclusion Criteria 1. Any condition or underlying disease that would pose an unacceptable safety risk or interfere with study assessments. 2. Use of other investigational drugs or specific prohibited medications outside of standard DMD care.

---

5. Intervention & Treatment Plan

5.1 Investigational Regimen Drug Name: Satralizumab **Trade Name:** Enspryng **ATC Code:** L04AC19 **EMA URL:** https://www.ema.europa.eu/en/documents/product-information/enspryng-epar-product-information_en.pdf **Dosage/Frequency:** Weight-tier based dose of satralizumab. Administered as loading doses on Day 1, Week 2, and Week 4, followed by maintenance doses every 4 weeks (Q4W) from Week 8 through Week 104. **Route of Administration:** Subcutaneous (SC) injection. **Duration of Treatment:** 104 weeks (2 years) (Truncated for early termination).

5.2 Concomitant & Prohibited Therapy Permitted: Standard daily corticosteroids commonly used to treat DMD. **Prohibited:** Concurrent experimental DMD therapies or unapproved biologics.

---

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	2-4 weeks	Informed Consent, Medical History, Fracture Assessment
Baseline	Day 1	First SC loading dose, DEXA scan
Loading	Weeks 2 & 4	Additional SC loading doses
Maintenance	Q4W (Weeks 8-104)	SC injection, safety monitoring, adverse event check-ups
End of Study	Week 52 / 104	Final DEXA scan, exit assessments (Currently adapted to a 6-month collection milestone)

---

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Change in lumbar spine Bone Mineral Density (BMD) Z-score from baseline to Week 52. **Measurement Method:** Dual-energy X-ray absorptiometry (DEXA) scan.

7.2 Secondary & Exploratory Endpoints - Incidence and average number of new broken bones (fractures). - Changes in muscle function and strength. - Pharmacokinetics (how the body processes satralizumab). - Safety and tolerability (number of participants experiencing unwanted effects).

---

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Regular check-ups every 4 weeks to monitor for unwanted effects. **Serious Adverse Events (SAEs):** Standard 24-hour reporting timeline. (Note: Roche clarified the study termination was *not* due to any new or unanticipated safety risks). **Data Monitoring Committee (DMC):** Supervised internally by Hoffmann-La Roche safety teams.

---

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) population and Safety Set for all participants who receive at least one dose. **Sample Size Justification:** Target $N=50$ adequately powered to assess the initial efficacy and safety signals of IL-6R inhibition in this rare disease population. **Handling of Missing Data:** Standard imputation methodology; interim analysis data will be shared with the community post-termination to advance bone health understanding in DMD.

---

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study is conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Regular onsite visits accompanied by continuous tracking of participant outcomes and safety. **Audit Trail:** Electronic Case Report Form (eCRF) locking and quality auditing for centralized DEXA scan readings.

---

11. Study Identifiers & Governance

Primary ID: BN45398 **Registry Number:** NCT06450639 **Posting ID:** 39366947587872783 **Sponsor/Lead Organization:** Roche (Hoffmann-La Roche) **Study Title:** SHIELD-DMD **Official Regulatory Title:** A Phase II Multicenter, Open-label Study to Evaluate the Efficacy, Safety, Pharmacokinetics, and Pharmacodynamics of Satralizumab in Pediatric Patients With Duchenne Muscular Dystrophy (SHIELD DMD)

---

12. Clinical Framework (DMD Specific)

Primary Condition / Disease Area: Duchenne Muscular Dystrophy (DMD) **Preferred UMLS Name:** Duchenne Muscular Dystrophy **MeSH Code:** D020388 **SNOMED ID:** 387732009 **Diagnosis Threshold:** Genetically confirmed DMD **Medical Methodology:** IL-6 Receptor Inhibition + Guideline Determined Corticosteroids **Optimization Target:** Bone mineral density preservation and fracture incidence reduction

---

13. Study Procedures & Methodology

Management Pathway: Standard DMD multidisciplinary clinical pathway integrated with IL-6R biologic therapy **Education Protocol (HCP):** Site training on weight-based SC injection protocols and standardized DEXA imaging **Education Protocol (Patient):** Caregiver education on bone health, fracture monitoring, and potential therapy side effects **Quality Audit Mechanism:** Centralized evaluation of DEXA scans and rigorous AE tracking

---

14. Observation & Follow-Up Schedule

Total Duration: 104 weeks (Truncated to a 6-month follow-up for enrolled participants) **Onsite Visit Cadence:** Every 4 weeks (Q4W) **Remote Monitoring Frequency:** Continuous safety reporting **Checklist Review Interval:** Every 12 weeks for major functional assessments

---

15. Sign-off & Approval

Role	Name	Signature	Date
Principal Investigator	[Name]	____	[DD-MMM-YYYY]
Clinical Operations Lead	**Dr. Levi Garraway**	___	[DD-MMM-YYYY]
Lead Statistician	[Name]	_____	[DD-MMM-YYYY]